

Select Forecast Horizon

7

Select Forecasting Model

Gradient Boosting

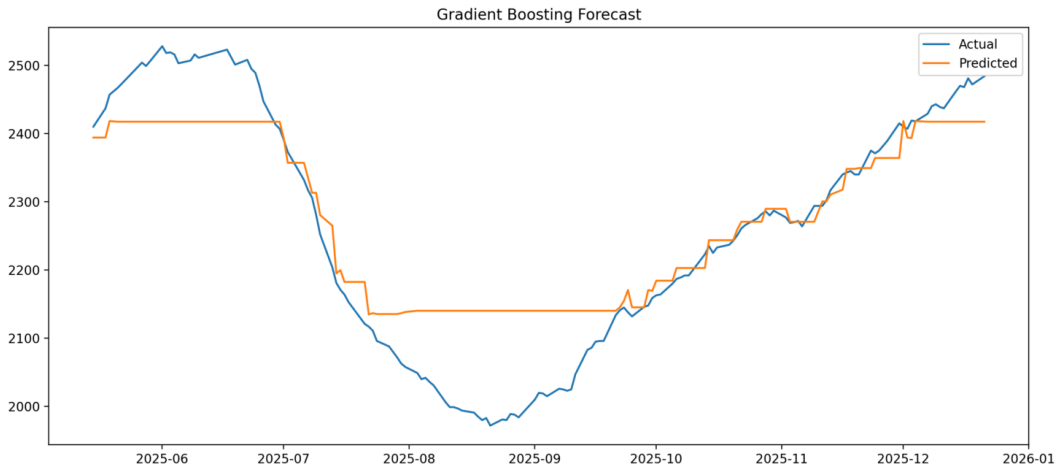
# Predictive Forecasting of HHS Care Load

Interactive forecasting dashboard for analyzing HHS Unaccompanied Children Program data.

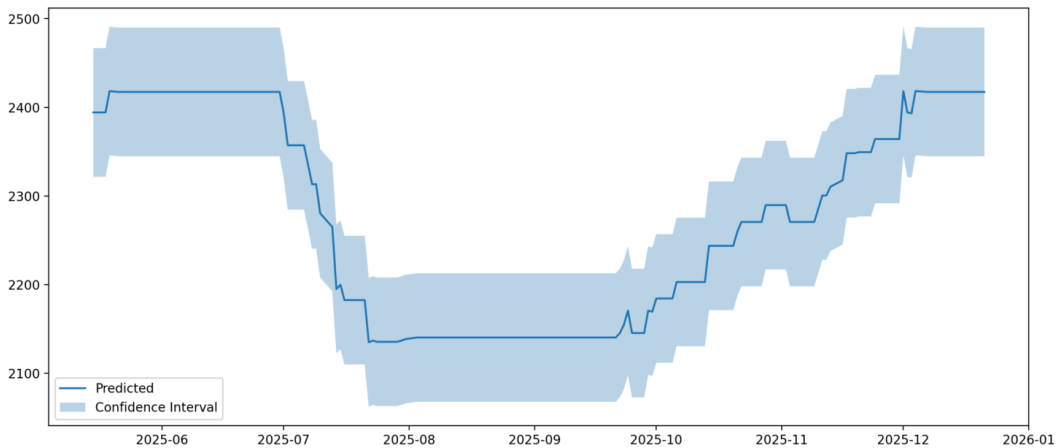
## KPI Summary

| Forecast Accuracy | MAE   | RMSE  | Average Care Load |
|-------------------|-------|-------|-------------------|
| 97.54%            | 53.27 | 72.50 | 6049              |

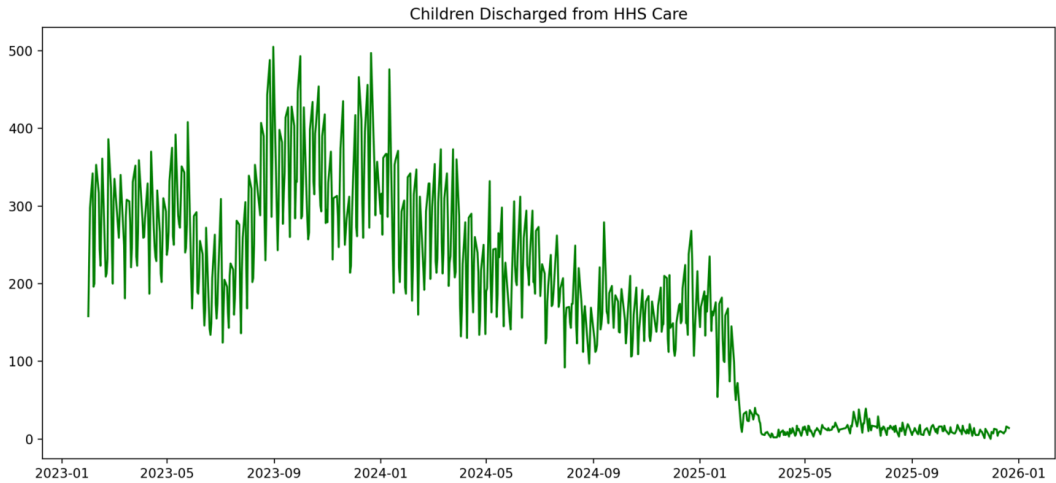
## Future Care Load Forecast



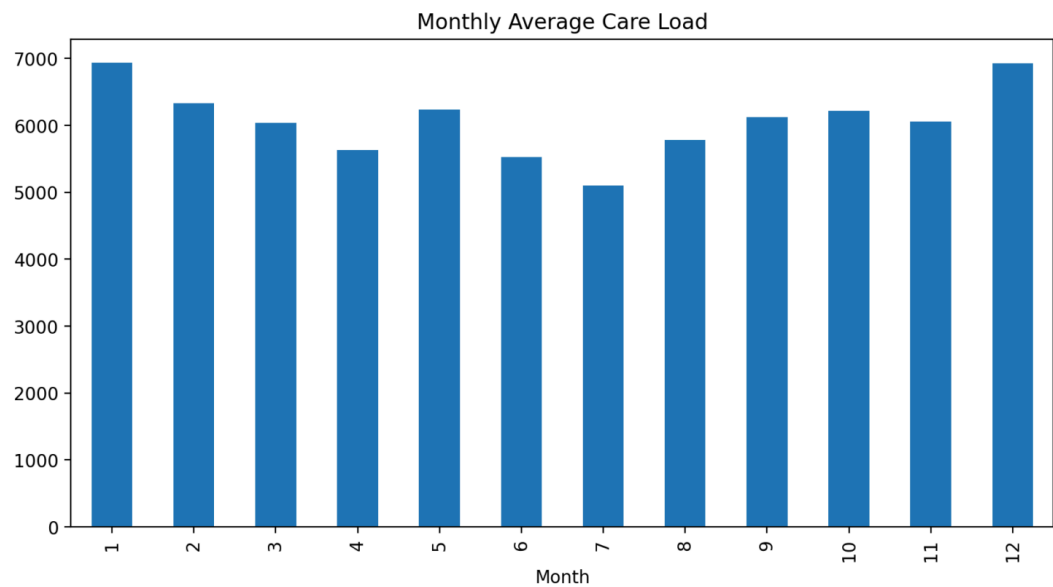
## Confidence Interval Visualization



## Discharge Demand Forecast



Monthly Trend Analysis



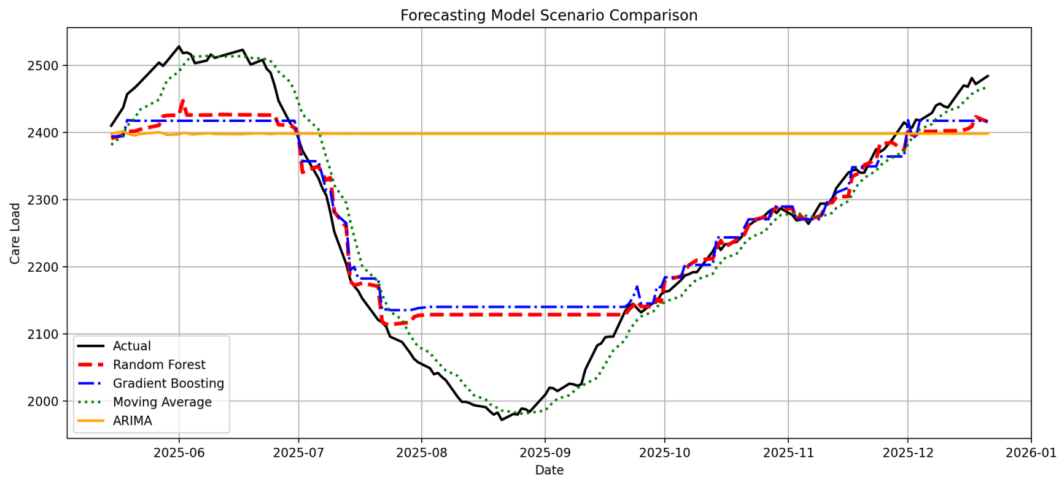
Scenario Comparison

|   | Actual | Random Forest | Gradient Boosting | Moving Average | ARIMA |
|---|--------|---------------|-------------------|----------------|-------|
| 0 | 2410   | 2390.94       | 2394.1731         | 2381.5714      |       |
| 1 | 2437   | 2396.28       | 2394.1731         | 2396.1429      |       |
| 2 | 2457   | 2401.49       | 2418.2893         | 2409.8571      |       |
| 3 | 2467   | 2401.83       | 2417.3654         | 2421.5714      |       |
| 4 | 2473   | 2404.13       | 2417.3654         | 2433.1429      |       |
| 5 | 2504   | 2410.76       | 2417.3654         | 2448.8571      |       |
| 6 | 2499   | 2423.94       | 2417.3654         | 2463.8571      |       |
| 7 | 2506   | 2425.26       | 2417.3654         | 2477.5714      |       |
| 8 | 2528   | 2425.8        | 2417.3654         | 2490.5714      |       |
| 9 | 2518   | 2447.2        | 2417.3654         | 2499.2857      |       |
|   |        |               |                   |                |       |

Model Comparison Metrics

| Model             | MAE      | RMSE     |
|-------------------|----------|----------|
| Random Forest     | 48.7400  | 66.7100  |
| Gradient Boosting | 53.2700  | 72.5000  |
| Moving Average    | 25.0900  | 30.8700  |
| ARIMA             | 189.5300 | 230.6300 |

Scenario Comparison Visualization



Dataset Preview

|     | Date                | Children apprehended and placed in CBP custody* | Children in CBP custody | Children transferred out of CBP custody | Children ir |
|-----|---------------------|-------------------------------------------------|-------------------------|-----------------------------------------|-------------|
| 712 | 2023-01-31 00:00:00 | 26                                              | 36                      | 36                                      |             |
| 711 | 2023-02-01 00:00:00 | 25                                              | 32                      | 27                                      |             |
| 710 | 2023-02-02 00:00:00 | 15                                              | 13                      | 23                                      |             |
| 709 | 2023-02-05 00:00:00 | 16                                              | 18                      | 15                                      |             |
| 708 | 2023-02-06 00:00:00 | 11                                              | 22                      | 12                                      |             |
| 707 | 2023-02-07 00:00:00 | 16                                              | 20                      | 20                                      |             |
| 706 | 2023-02-08 00:00:00 | 93                                              | 196                     | 169                                     |             |
| 705 | 2023-02-09 00:00:00 | 124                                             | 234                     | 161                                     |             |
| 704 | 2023-02-12 00:00:00 | 92                                              | 203                     | 173                                     |             |
| 703 | 2023-02-13 00:00:00 | 186                                             | 259                     | 172                                     |             |
|     |                     |                                                 |                         |                                         |             |

Download Cleaned Dataset